

Breast Cancer T-Stage Prediction and Classification Using Machine Learning

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Abstract—Routine early breast cancer screenings through mammography are now common across the globe. As screening rates continue to rise, accurate staging of breast tumors plays a critical role in determining prognosis, treatment planning, and long-term patient outcomes. Thus, we develop a machine learning framework for automated prediction of the T-stage (tumor-size stage) using a combination of clinical data and quantitative MRI-derived radiomic features. Our work integrates a preprocessed multimodal dataset that provides clean, standardized inputs for model training. We train four supervised learning models on both the full feature set and a reduced feature subset to evaluate algorithmic behavior under different levels of feature complexity. This work establishes a reliable baseline framework for automated T-stage prediction and provides a foundation for future work involving model calibration, multimodal integration, cross-validation, and the development of more robust architectures for clinical deployment.

I. INTRODUCTION

Medical professionals must first determine the correct stage of a breast cancer patient by examining numerous important clinical and imaging features. Their decisions are crucial for determining the most effective treatment plan. The primary tumor size (T-stage) is an essential component of cancer staging [5]. Although radiologists can manually interpret MRI scans and clinical information to determine the stage of the cancer, the process is time-consuming, subjective, and prone to interpretive bias.

Machine learning has evolved into a promising tool for assisting health professionals by providing quick, consistent, data-driven predictions [11]. Machine learning models can generate a comprehensive view of tumor characteristics by combining a large number of MRI-derived features with a cleaned clinical dataset, thereby providing highly accurate predictions of cancer stage.

In this work, we develop a supervised learning framework to predict tumor T-stage using a cleaned dataset of clinical information and radiomic features extracted from breast MRI scans. The focus of this work is to construct a reliable preprocessing pipeline, prepare clean data for multiple model training, and analyze the results to refine the model for optimal prediction performance. The goal is not only to generate predictive results but also to establish a structured foundation for later model refinement, hyperparameter tuning, and multimodal experimentation.

II. BACKGROUND

Magnetic Resonance Imaging (MRI) is one of the most important tools for studying breast cancer because it provides detailed images of soft tissue. Unlike X-ray or CT imaging, MRI does not use ionizing radiation. It relies on a strong magnetic field and radio waves to measure how water molecules in tissue respond. This makes MRI effective for identifying tumor characteristics, as cancerous tissue often shows different structural and vascular behavior compared to healthy tissue.

In breast cancer applications, dynamic contrast-enhanced MRI (DCE-MRI) is especially valuable [8]. Tumors absorb and release contrast agents differently from normal tissue, which helps highlight tumor boundaries, vascular activity, and overall morphology. These contrast patterns give clinicians detailed information about tumor progression and are widely used to evaluate cancer stage.

Radiomics expands the use of MRI by converting medical images into quantitative features [6]. These features describe texture, intensity patterns, shape, and how the tumor enhances after contrast injection [7]. In this work, radiomic features provide the model with a structured representation of each tumor without requiring manual inspection of every MRI volume.

Clinical information adds another important layer of data. Variables such as patient age, receptor status, pathology results, and treatment history influence how breast cancer behaves and how it should be classified. Imaging shows what the tumor looks like, but clinical variables provide biological and medical context. Combining these two sources gives the model a fuller understanding of each case.

Multimodal learning brings these modalities together in a single system. In medical applications, combining MRI features with clinical information often improves prediction accuracy because each data type contributes different but complementary details [2]. This work uses this approach to build a stronger input representation for the classification of tumor stages.

Convolutional neural networks (CNNs) form another essential component of this background. CNNs have long been the dominant architecture in medical image analysis due to their ability to extract hierarchical spatial features through

convolutional layers. Modern CNN backbones such as ResNet introduce residual connections that enable deeper networks to be trained efficiently while preserving gradient flow. These architectures excel at capturing both low-level patterns and more complex anatomical structures, making them well suited for an MRI-based breast cancer assessment. However, traditional 2D CNNs process each slice independently and therefore fail to fully exploit the three-dimensional nature of MRI or CT volumes. This is an important limitation as tumors often extend across multiple slices. This motivates the use of our 2.5D CNN approach described next, which integrates information across slices while retaining the efficiency and representational strength of a ResNet-based convolutional backbone.

A. 2.5-Dimensional CNN

The 2.5D CNN refers to a practical compromise between 2D and 3D convolutional neural networks, designed to balance computational efficiency with volumetric awareness. In medical imaging, where CT and MRI scans are inherently 3-dimensional, this balance is crucial. 2D CNNs are fast, lightweight, easy to train, and are strong at preserving high spatial resolution. Their limitation, however, is that they operate on singular slices and therefore have no understanding of volumetric context. 3D CNNs, in contrast, capture full volumetric structure by applying 3D kernels, providing strong performance for tasks requiring inter-slice reasoning. Nonetheless, 3D CNNs are memory-intensive, computationally expensive, slower to train, and often require downsampling, which reduces spatial resolution. The term 2.5D CNN, standardized formally in publications around 2018, emerged as a middle ground. The core idea is that we stack a small amount of neighboring slices (3-7) along the channel dimension. This strategy transforms 3D structural information while retaining a 2D image's spatial shape. This approach allows us to train a limited 3D volume with a 2D CNN. Therefore, we gain the following benefits:

- Stronger use of volumetric context compared to standard 2D CNN.
- Compatibility with pretrained 2D CNN backbones.
- Computational efficiency.
- Data efficiency.

This research centers on classification and its applications in multimodal learning.

B. Multimodality

Multimodality refers to neural networks that simultaneously learn multiple data types. In our case, we leverage a fusion of clinical tabular data and medical imaging data to produce informed predictions of a patient's tumor stage. It is critical to keep in mind that, "While the fusion approach varied widely, the consistent improvement in reported performance across a wide variety of clinical use cases suggests that model performance based on single-modal data may not represent state of the art for a given application when multimodal data are not considered". Adopting multimodal approaches in healthcare is crucial for advancing medical technology.

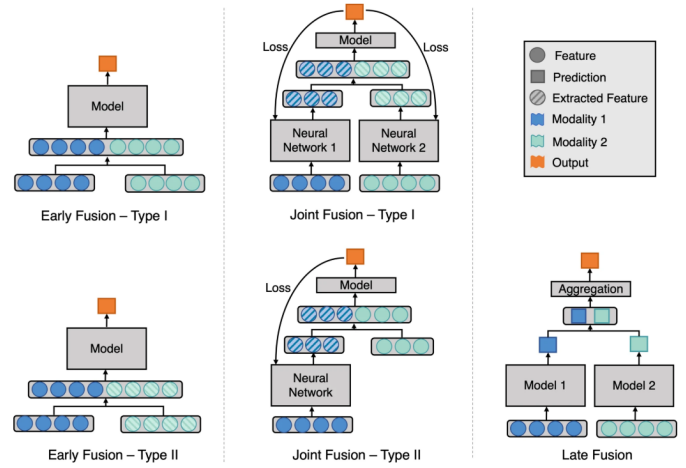


Fig. 1. Fusion strategies using deep learning. Model architecture for different fusion strategies. Early fusion (left) concatenates original or extracted features at the input level. Joint fusion (middle) also joins features at the input level, but the loss is propagated back to the feature-extracting model. Late fusion (right) aggregates predictions at the decision level. Adapted from: "Fusion of medical imaging and electronic health records using deep learning: a systematic review and implementation guidelines," npj Digital Medicine.

III. DATASET AND DATA PREPROCESSING

A. Dataset

The dataset used for this work is available on The Cancer Imaging Archive (TCIA) and contains biopsy-proven invasive breast cancer images collected by the Duke Breast Cancer MRI Collection [1]. The collection includes images from a total of 922 patients, spanning from January 2000 to March 2014. Each patient record contains multiple MRI sequences, including pre-contrast, first post-contrast, and subtracted images, along with annotation boxes drawn by fellowship-trained breast radiologists. Each patient typically has 4 to 7 MRI sequences, totaling 5,161 sequences. The resulting image slices total about 773,253 DICOM MRI slices.

In addition, TCIA provides a comprehensive dataset of clinical data gathered from medical records and radiology reports. These include features such as receptor status (ER/PR/HER2), tumor grade, molecular subtype, staging information (T, N, M components), tumor location, lymph node involvement, patient age, menopausal status, and several mammography and ultrasound descriptors [1].

For this work, we select the TCIA "Clinical and Other Features" dataset with 90 features and "Imaging Features", which contain 529 radiomic features extracted from tumor regions and fibroglandular tissue. These features include measurements of tumor size, morphology, enhancement kinetics, texture, spatial heterogeneity, and FGT characteristics. After combining the two datasets through the preprocessing pipeline, the cleaned clinical features are merged with the imaging features to create the final modeling dataset stored as `final_features.csv`, which serves as the foundation for all experiments in this study.

B. Feature Processing

The objective of our feature-processing pipeline is to reduce the total number of features while preserving their relevance and correlation to the target labels. The pipeline consists of the following steps:

- **Feature Merging:** We merge clinical features with image-derived features using patient ID as the key. This produces a unified dataset containing both clinical and imaging information for each patient.
- **Handling Missing Values:** Features with excessive missing values above a specific threshold are flagged for removal. For features with only minimal missing values, we impute missing entries using the median to avoid introducing bias from outliers.
- **Feature Selection:** Although the dataset is cleaner, it still contains several hundred features. To further reduce dimensionality, we:
 - Remove highly correlated features that contribute little additional variance.
 - Apply a Random Forest model to identify and discard features exhibiting low importance.
 - Calculate Pearson correlations with our three target labels (TNM) and retain only the features most strongly correlated with them.
 - Principal Component Analysis (PCA): We apply PCA to specific subsets of features that exhibit strong internal correlations. This allows us to combine redundant variables and further reduce dimensionality while retaining variance.

After completing the preprocessing pipeline, we obtain a final set of 19 significant features that provide meaningful and compact context for each patient.

C. Image Processing

The goal of the image processing pipeline is to load and standardize 3-dimensional MRI volumes, each consisting of a variable number of slices. The MONAI framework provides a robust set of tools for processing medical imaging [12]. All image volumes are loaded in batches and passed through a sequence of MONAI transformations to ensure consistent spatial orientation, intensity scaling, and input dimensionality. The transformations applied are as follows:

- **Crop to region of interest (ROI)** - crops full sequence into an ROI containing crucial tumor features utilizing annotation boxes.
- **Spacing (Resampling)** - All volumes are resampled to a uniform voxel spacing to account for differences across scans. Standardizing the physical resolution ensures that spatial features are comparable across subjects and prevents the model from learning scanner-dependent features.
- **After resampling, volumes are resized to a fixed spatial dimension.** Padding is applied when images are smaller than the desired size, and cropping is used when they are

larger. This generates consistent input dimensions for the network while minimizing distortion.

- **Z-Score Standardization** - Intensity values are normalized per volume by subtracting the mean and dividing by the standard deviation. According to Reinhold et al. (2019), “intensity normalization as a preprocessing step improves the synthesis results across all investigated synthesis algorithms” [4]. This standardization reduces intensity variability across scans and enhances model performance.
- **Min-Max Normalization** - Scales image pixel values to be between 0 and 1. This brings all image features to a uniform numeric scale, facilitating faster and more stable model training by reducing numerical instability and mitigating vanishing or exploding gradients.

These augmentations assist in building a consistent processing and training pipeline to boost model efficiency and performance.

IV. METHODOLOGY

This section will outline our methodology for manipulating the data for TNM staging classifications. We explore multiple algorithms, comparing performance along a variety of metrics such as accuracy, recall, and f1-score.

A. Classifiers

Following our feature and image preprocessing, we implement four distinct algorithms using a subset of our clinical feature dataset.

- **Random Forest** - Random Forest is a commonly used application for biomarker discovery or clinical data analysis due to its robustness to noisy and high-dimensional data.
- **Support Vector Machine (SVM)** - SVMs are often seen in clinical scenarios, primarily for classification, diagnosis, and prognosis. Support Vector Machines perform well with high-dimensional but small to medium-sized datasets.
- **K-Nearest Neighbors (KNN)** - KNN serves as a simpler baseline algorithm for comparison. KNN is frequently seen in disease classification for small datasets. However, performance worsens with noisy and high-dimensional data; thus, we employ a smaller subset of our clinical feature data.
- **2.5-Dimensional CNN** - Lastly, we leverage a multimodal 2.5D CNN to take advantage of imaging and clinical data. Our approach uses late fusion to concatenate all the data into one final multi-layer perceptron (MLP). In other words, the imaging data is transfigured through a convolutional architecture, and the clinical features are simultaneously transformed through an artificial neural network to ultimately receive two outputs. Subsequently, these results are concatenated into the last MLP to produce a T-stage prediction. The T-stage corresponds to tumor size and represents the internationally recognized standard for classifying the anatomical extent of malignant tumor spread.

V. RESULTS

This section presents the performance evaluation of our models across clinical and radiomic data modalities. We first examine performance across our traditional machine learning models, followed by an analysis of multimodal integration networks.

A. Imaging Features and Clinical Pipeline Results

We evaluated four models for the clinical and imaging feature pipeline: Random Forest, Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbor (KNN). Random Forest achieved the strongest overall performance across accuracy, precision, and recall and showed the clearest separation of early-stage tumor classes, as shown in Figure 2.

Logistic Regression produced moderate results and struggled with later tumor stages due to the nonlinear structure of the feature space. SVM showed inconsistent performance with overlapping decision boundaries, particularly in advanced stages. KNN performed the weakest because the tumor stages did not cluster cleanly in the feature space, which limited its ability to form meaningful neighborhoods.

Because SVM and KNN are sensitive to scaling, we applied a Standard Scaler to evaluate potential improvements. SVM showed a slight accuracy increase from 0.58 to 0.61, while both SVM and KNN experienced drops in precision and recall. This indicated that scaling alone did not address the challenges presented by later-stage cases.

Across all models, predictions for early stages were consistently stronger than predictions for later stages. This is visible in the confusion matrices, where stages 0 and 1 appear concentrated along the diagonal, while stages 2 and 3 are more diffuse. Figure 3 highlights this pattern for Random Forest and shows how limited late-stage representation and similarity make advanced cases harder to classify.

Overall, Random Forest was the most effective model for this pipeline. The decline in performance for later-stage predictions across all models demonstrates the need for more balanced and diverse data to improve classification of advanced cases.

B. 2.5-Dimensional CNN

This section evaluates the performance of our multimodal 2.5D CNN and provides insight into our model's reasoning process.

1) *Training Dynamics*: Figure 4 illustrates the training and validation loss curves throughout the model training process. The convergence patterns demonstrate stable learning with minimal overfitting, achieving a final loss of approximately 0.4 and convergence at around 30 epochs. The close alignment between training and validation curves indicates good generalization capability without significant overfitting.

2) *Classification Performance*: Table I presents comprehensive classification metrics including precision, recall, and F1-score for each T-stage class. The model achieved an overall accuracy of 68%, with weighted average precision, recall, and F1-scores of 0.70, 0.68, and 0.68, respectively. Performance

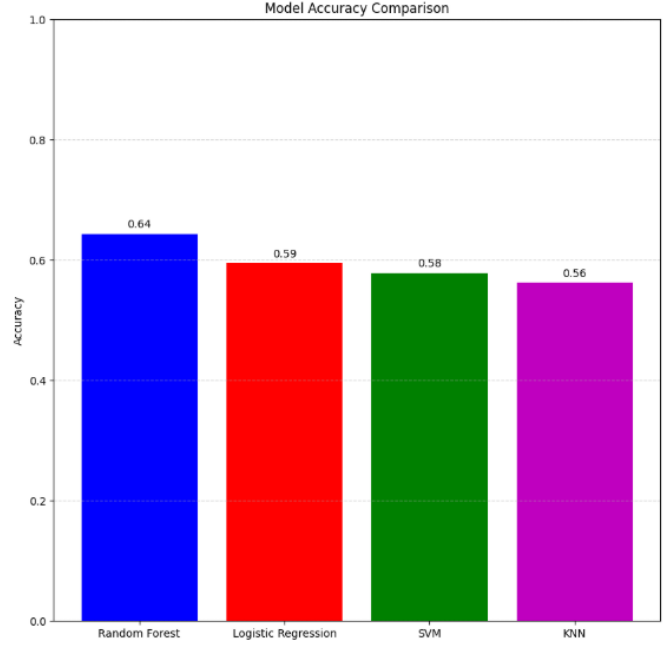


Fig. 2. Strongest accuracy performance for Random Forest, while the remaining models show reduced performance and greater sensitivity to feature complexity.

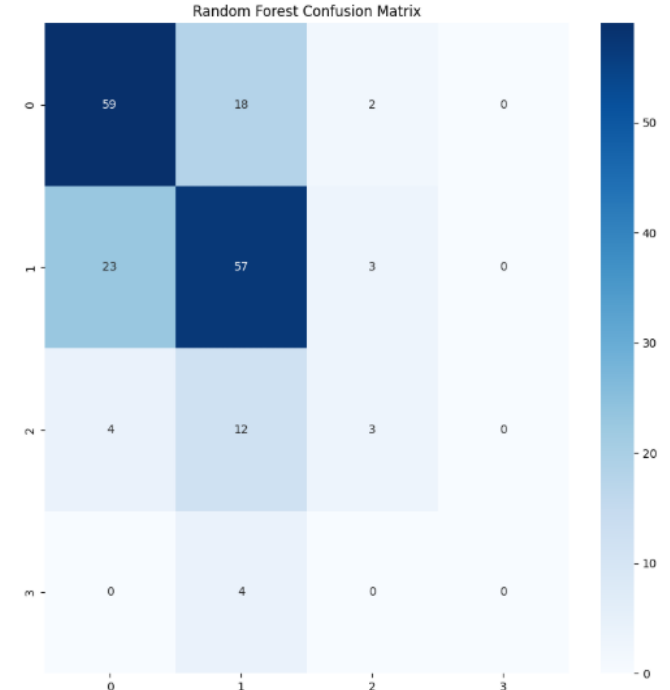


Fig. 3. Confusion matrix for the Random Forest model. High true-positive counts in stages 0 and 1 highlight the model's ability to capture early-stage features. In contrast, later stages show dispersed and low-frequency predictions, consistent with class imbalance and limited distinguishable imaging or clinical signatures in advanced disease cases. This reflects the need for broader and more diverse late-stage samples.

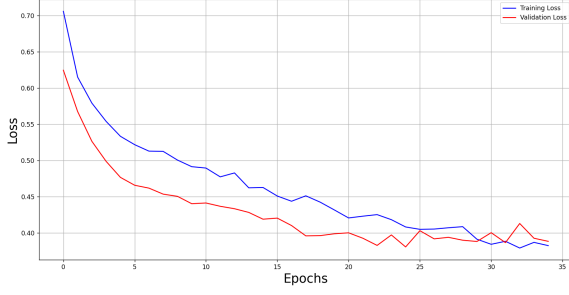


Fig. 4. Training and validation loss curves over 35 epochs.

varied considerably across T-stages, with the model demonstrating exceptional performance on T4 tumors (F1-score: 0.97, precision: 0.94, recall: 1.00) but more modest results for earlier stages. T1 tumors proved most challenging (F1-score: 0.62), likely due to their subtle radiological features and smaller tissue volumes. The high recall for T3 tumors (0.87) suggests the model effectively identifies these intermediate-stage cases, though with somewhat lower precision (0.63).

Class	Precision	Recall	F1-Score	Support
T-Stage 1	0.75	0.54	0.62	84
T-Stage 2	0.62	0.65	0.64	75
T-Stage 3	0.63	0.87	0.73	52
T-Stage 4	0.94	1.00	0.97	17
Accuracy			0.68	228
Macro Avg	0.74	0.76	0.74	228
Weighted Avg	0.70	0.68	0.68	228

TABLE I
CLASSIFICATION METRICS FOR T-STAGE PREDICTION.

3) *Error Analysis*: The confusion matrix (Figure 5) reveals the distribution of correct classifications and misclassifications across T-stages. Notable confusion patterns include frequent misclassification between lower tumor sizes, particularly T1 and T2 stages, suggesting overlapping features across adjacent tumor stages. In contrast, the model performed strongest for T4 tumors, with only one missed prediction out of 17 cases. This superior performance on later-stage tumors reflects their more distinctive morphological characteristics, including larger size, irregular edges, and invasion of surrounding structures.

4) *Model Interpretability*: Explainable AI (XAI) is essential not only for validating that the model’s decision-making process reflects clinically relevant features but also for building clinician trust in AI-assisted diagnostic systems. Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to visualize the spatial regions that most strongly influenced model predictions. Figure 6 represents a correct T-stage 4 prediction. The activation map demonstrates the model’s attention to the tumor boundaries and surrounding tissue. The strong performance on large tumors is attributed to their more distinctive edges and textural features, which provide clearer discriminative patterns compared to earlier-stage tumors. These visualization results confirm that the

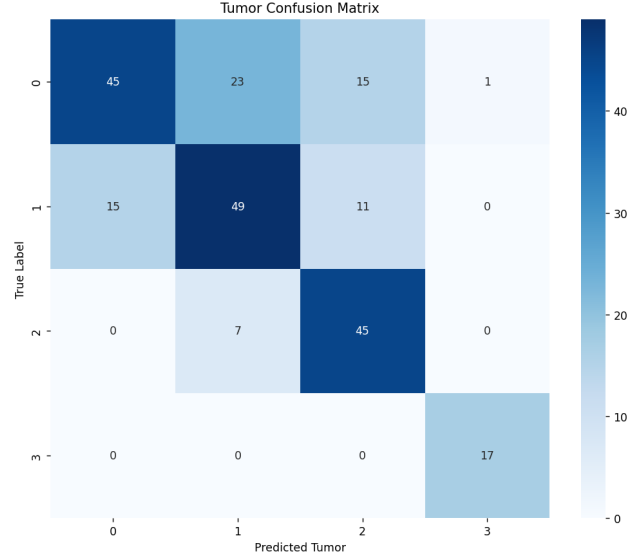


Fig. 5. Confusion matrix displaying predicted versus actual T-stage classifications.

model focuses on clinically meaningful regions rather than irrelevant image properties, supporting its potential utility in clinical workflows.

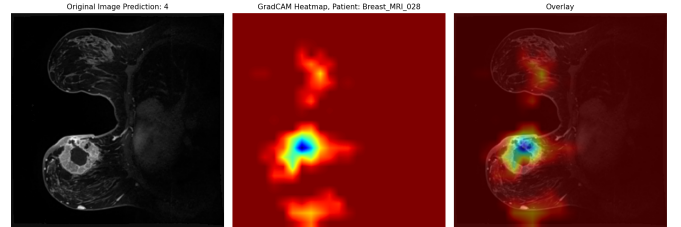


Fig. 6. Grad-CAM visualization for a correctly predicted T4 tumor.

VI. CONCLUSION

The main goal of this research is to build a machine learning framework capable of predicting breast cancer T-stage using clinical features and MRI-derived imaging data. The work completed so far establishes the preprocessing pipeline, integrates the datasets, and prepares the models for supervised learning. Although the final evaluation metrics are not yet available, the current progress confirms that the system is ready for full training and analysis. These foundations place the work in a strong position to complete the remaining phases, including running full training, collecting performance metrics, and evaluating how well each model predicts tumor stage. Based on the work completed, the work is on track to meet its goal once the modeling and evaluation phase is finalized.

VII. FUTURE WORK

Our work has numerous possibilities for expansion and improvement. Future directions for this work may include incorporating tumor analysis based on tumor grading through biopsy datasets [10]. Aggregating staging and grading results for a patient's tumor could significantly assist in determining treatment plans and prognosis. Additionally, to further advance our model's practicality, we could translate our Tumor, Node, Metastasis predictions into a final tumor stage prognosis ranging from stage I–IV. The system can increase the dataset size and diversity to make it more robust. Incorporating data from other institutions and locations, as well as more advanced images and patient demographics, would help the model generalize more effectively. Additionally, integrating the system into the clinical workflow environment allows for valuable insights. Radiologists and other health professionals could interact with the system and provide meaningful feedback. This will improve usability, identify weaknesses, and help determine whether the system's predictions align with real-world decision-making. Consequently, this process would help the current model overcome its limitations as merely an academic prototype and develop into a more reliable decision-support tool.

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